BUSINESS PROPOSAL / CWC-FIN-01

Financing American canals through investable returns

Continental Waterways Construction | Proposed PPP and concession financing framework



IRR-led appraisal. Explicit public support. Disciplined project selection.

A business proposal for a first 10-mile canal corridor, using the ACN-100 standard developed in this series. The proposal links construction budgets, freight tariffs, debt coverage and shareholder returns to a transparent public-private funding structure.

ILLUSTRATIVE BASE CASE	MODEL RESULT
Total financing requirement	$3,425.44m / $3.425bn
Private equity hurdle	12.00% nominal, pre-tax; not a promised return
Public construction contribution	$2.000bn; no grant award assumed
Opening annual availability payment	$96.16m; escalates at 2% per year

Independent fictional sponsor concept. No canal route, concession, investment product, government contribution or loan is approved. Costs, demand, rates and returns are analyst assumptions. IRR is a test of cash flows, not a source of money.

01 / INVESTMENT RECOMMENDATION

A supported PPP, not a toll-only financing claim

Form a corridor-specific concession company to design, build, finance, operate and maintain a useful connection between existing freight markets. CWC is the proposed development and delivery sponsor; construction and operations must be competitively procured. Infrastructure investors supply equity, lenders supply debt and a public authority supplies explicitly budgeted support.

Decision metric	Illustrative result
Funding need	$3,425.44m including construction, additions, finance costs and liquidity
Capital structure	$2,000m public contribution + $800m senior debt + $625.44m equity
Equity return	12.00% nominal pre-tax over five construction and forty operating years
Gross asset return with availability payments	3.83%; before grant, debt and financing costs
Merchant asset return without public payments	-2.09%; excludes capital grant and availability payments
Minimum base-case DSCR	1.74x; cash flow available for debt service / annual debt service

The 12% equity IRR is reached by solving for the public availability payment. It is not independently demonstrated demand or a market-clearing tariff. The public contribution and payments explain the difference between a low asset return and a higher private equity return.

Proceed only to feasibility and commercial testing. Reject or redesign a corridor if shippers will not contract for the service, the public authority cannot afford its obligations, or an independent public-value appraisal favors another mode.

02 / INVESTABLE PROJECT SCOPE

Start with a functioning 10-mile connection

The worked case is a hypothetical rural corridor: 10 open-canal miles, two automated locks, one bridge and a modest freight-terminal package. It has no selected geographic alignment. A short canal creates value only if it connects meaningful origins and destinations; a disconnected demonstration waterway cannot support the assumed traffic.

Scope	Planning assumption
Navigation standard	ACN-100: 70 m bed width; 4.5 m maintained depth
Locks	Two usable 110 m x 18 m chambers; 6 m planning lift each
Bridge	100 m clear opening; 12 m navigation headroom at high navigable water
Fleet interface	Individual freight vessels up to 80 m x 15 m; coupled units up to 100 m overall
Terminals	$150m new provisional capital allowance, additional to the preceding corridor illustration
Operating term	40 years after five construction years; no terminal sale value
Water / permissions	No water entitlement, land agreement, bridge approval or environmental authorization assumed

The dimensions follow the preceding American Canal Network proposal. They are proposed planning values, not adopted U.S. regulations or proof of construction feasibility. Regional river-tow routes may need larger structures. [1]

Do not roll Arizona tunnels or Rocky Mountain lock flights into this model. Those have different capital, hydrological, ecological and operating risks and require separate investment cases.

03 / COST BUILD-UP

From corridor illustration to funding requirement

Use of funds	US$m	Basis
Open canal: 10 x $150m	1,500.00	Prior illustrative ordinary-ground rate
Locks: 2 x $250m	500.00	Prior illustrative rate; not a supplier quote
Bridge: 1 x $60m	60.00	Prior illustrative rate
Design, programme and contingency	721.00	35% of the $2,060m direct corridor works
Freight terminal package	150.00	Additional provisional allowance
Land, water, environmental / utility provision	200.00	Placeholder allowance; not secured rights or a valuation
Additional inflation-risk provision	100.00	Nominal construction budget allowance
Construction interest	103.40	5.5% on $800m staged draws; midyear convention
Debt arrangement fee	16.00	Assumed 2% of senior commitment
Debt-service reserve	55.04	One base-case annual debt-service payment
Working liquidity	20.00	Funded at construction completion
TOTAL	3,425.44	Annual model uses unrounded values

The preceding $2.78bn corridor illustration excluded several project costs. This proposal adds explicit allowances; it is not an unchanged quote. The $200m site provision may be insufficient and does not secure a water supply. The $100m inflation provision is not a detailed escalated estimate. Taxes and exceptional unknown costs require separate resolution.

The 35% allowance relates only to direct corridor works. It is not a statistically derived contingency or insurance against every overrun. A quantity surveyor and independent engineer must replace the budget before financing.

04 / SOURCES AND FUNDING MECHANICS

Who contributes the cash, and when

Capital source	US$m	Share	Proposed role
Public capital contribution	2,000.00	58.4%	Milestone-based capital grant / public capital funding
Senior secured debt	800.00	23.4%	Construction draws, then 30-year amortization
Private equity	625.44	18.3%	Sponsor and infrastructure investor capital at risk
TOTAL	3425.44	100.0%	No funding commitment exists

Debt and public draws follow 10%, 20%, 30%, 25% and 15% annual profiles. Equity pays the residual annual uses, including fees, interest and closing liquidity. This is a planning draw schedule, not an approved equity-first / pro-rata lender arrangement.

Build year	Uses	Debt	Public	Equity
1	341.30	80.00	200.00	61.30
2	655.00	160.00	400.00	95.00
3	989.10	240.00	600.00	149.10
4	839.65	200.00	500.00	139.65
5	600.39	120.00	300.00	180.39

All amounts above are US$m. Annual sources equal uses. Construction interest is paid as a project use and financed within this same funding package; it is not added again to the $800m loan principal. [Model]

Public support must have an identified lawful funding source and enforceable timing. An assumed grant is not cash. Do not begin major works with a gap disguised as a future federal award.

05 / COMMERCIAL REVENUES

Underwrite freight first; treat leisure as ancillary

Revenue input	Opening-price assumption
Steady freight volume	12 million U.S. short tons per year; not metric tonnes
Canal toll	$4.00 per short ton: $48m at steady volume
Terminal handling	$2.00 per short ton: $24m at steady volume
Ancillary revenue	$6m per year at steady level: marina / property / visitor concessions
Commercial ramp	60%, 80%, 90%, 100% in operating years 1-4
Escalation	2% annually from first operations on unit tariffs and ancillary rates
Operating costs	$35m in operating year 1; grows 2% annually
Renewal provision	$12m in operating year 1; grows 2%; treated as spent / committed each year

The $78m steady-volume commercial revenue is expressed at opening prices. Year 4 nominal commercial receipts are $82.77m after escalation. Availability payments are additional and separate; no grant is counted as operating income.

Demand evidence required

Secure independently reviewed shipper origin-destination demand, comparative door-to-door costs, port and lock capacity, and binding minimum-volume or take-or-pay contracts where commercially justified. Contracted revenue remains exposed to counterparty default, service failures and contract expiry.

At 2,000 short tons per laden trip, 12m short tons means 6,000 laden movements annually, or about 16.4 per day. An empty return for each requires roughly 32.9 freight movements daily before passenger traffic. Each lock, approach and terminal must support that demand; capacity is not yet demonstrated.

06 / RETURN METRICS

Use IRR with NPV, debt coverage and public value

Internal rate of return: the discount rate r for which the discounted cash flows sum to zero. It measures a return implied by specified cash flows; it does not create revenue, fund a capital gap or establish that a project is the best public investment. OMB guidance also treats IRR as supplementary information rather than a sufficient decision rule. [2]

Metric	Cash-flow definition in this proposal
Merchant asset IRR	Negative non-financing construction / working capital; positive commercial cash flow after operating and renewal costs; no public funding.
Supported gross asset IRR	Same asset cash flows plus availability payments; excludes capital grant, debt, financing fees and DSRA.
Private equity IRR	Equity contributions during construction and any stress support, followed by cash distributions after debt and reserve rules. Includes the benefit of public support.
NPV	Present value at an explicitly selected hurdle. Equity NPV at 12% is zero by construction in the solved base case.
DSCR	Cash flow available for debt service (CFADS) divided by interest plus principal due in the year.

Calculation convention

Time 0 is financial close, with no modeled flow. Contributions occur in years 1-5; operations in years 6-45. IRRs use annual end-year cash flows, not exact-date XIRR. Construction interest alone uses assumed midyear draws. All quoted returns are nominal, pre-tax, before investor-level fees.

State, federal and local taxes, depreciation, tax credits and each investor’s tax position are not modeled. The 12% hurdle is an analyst-selected commercial test, not an investor commitment or an after-tax return. Negative asset IRR is economically meaningful here, not a formatting error.

07 / BASE-CASE FINANCIAL RESULTS

A 12% equity return requires public payments

Output	Calculated result
Required opening availability payment	$96.16m per year, escalated 2% annually
Private equity IRR	12.00%
Supported gross asset IRR	3.83%
Merchant asset IRR without public funding	-2.09%
Gross asset NPV at illustrative 7% nominal	$-1,122.24m
Annual senior debt service	$55.04m
Minimum base-case DSCR	1.74x in first operations
Equity distributed in first operating year	$40.91m

These are separate viewpoints

The 3.83% asset IRR includes availability payments, but does not subtract the public capital grant from asset cost. The 12% equity return applies to only the private equity contribution after public grant and debt funding. Do not represent the 12% as a return on the entire canal or as a return earned by the taxpayer.

Without availability payments, first-year commercial revenue is $46.8m against $47m of operating and renewal provision. CFADS is negative $0.2m, before $55.04m of debt service. A capital grant alone does not make this debt service affordable during the assumed ramp.

Even with public payments, the gross asset NPV at the illustrative 7% nominal financial test is negative $1.12bn. That rate is not asserted to be the government’s mandated social discount rate. An independent public appraisal must quantify genuine benefits and compare alternative delivery choices.

08 / AVAILABILITY PAYMENT AND FISCAL IMPACT

Show the public obligation in full

Propose a performance-based annual payment for an available navigation corridor, working locks, verified depth and maintained assets. The private SPV retains modeled toll and terminal revenue. Any such hybrid arrangement needs a statutory authority, procurement, appropriation treatment and public affordability approval. It is not a sovereign guarantee.

Public commitment	Model result
Construction contribution	$2.000bn over years 1-5
First annual availability payment	$96.16m
Operating year 40 payment	$208.16m
40-year availability payments, undiscounted	$5.808bn
Construction + availability, undiscounted	$7.808bn
PV at assumed 7% nominal from financial close	$2.790bn
Public upside or termination receipts	None assumed in these totals

These figures exclude the authority’s own administration, tax effects and any extra overrun or termination exposure. Discounted costs are not comparable to undiscounted totals without stating the basis. Payments are not netted against monetized social benefits here.

Recommended public protections

Apply availability deductions for closures and performance failures; cap indexation as negotiated; share refinancing gains; audit related-party contracts; set a handback condition and renewal schedule; and consider profit / traffic upside sharing. None of that upside is assumed in the 12% model. Adding a sharing mechanism requires repricing the payment and equity case.

The modeled opening payment is solved from investor cash flows. A real authority should procure a competitive price and assess value for money, rather than accept a sponsor’s IRR target as a payment entitlement.

09 / SENIOR DEBT AND CASH CONTROLS

Debt must be repaid from operating cash

Proposed financing term	Illustrative assumption
Commitment / operations opening balance	$800m; no guaranteed placement or rating
Construction interest	5.5% on staged draws; $103.4m modeled total
Operating coupon	5.5% fixed assumption; hedge / rate lock not obtained
Amortization	30 equal annual payments from operating year 1
Annual debt service	$55.04m
Opening DSRA	$55.04m; released after scheduled repayment in operating year 30
Distribution threshold	1.30x annual DSCR; trap positive residual cash below threshold
Loan security	Proposed pledge over permitted concession rights, accounts and contractual receivables; subject to law
Government credit / tax exemption	None assumed

Cash waterfall

Pay essential operating costs and committed lifecycle expenditure first; pay senior interest and principal; maintain required liquidity and cure deficiencies; then distribute permitted residual cash. The simplified model traps positive distributions when DSCR is below 1.30x and releases trapped cash when that test is restored.

A 1.30x-only test needs an opening availability payment of about $71.76m under the base operating assumptions. That is less than the $96.16m payment needed to reach the selected equity hurdle. Lender coverage alone therefore does not establish equity investability.

The model assumes no debt refinancing proceeds and no balloon repayment. If operations cannot cover scheduled debt, reserve cash is finite; any sponsor support or restructuring must be explicitly funded. Loan-life coverage, liquidity stress and covenant definitions require a lender model.

10 / PPP DELIVERY STRUCTURE

Separate the builder, asset company and authority

Participant	Proposed responsibilities
Public authority	Own or control permitted corridor rights; approve public expenditure; procure concession; oversee performance and public access.
Concession SPV	Hold the concession; receive tolls and payments; borrow; pay construction and operating contractors; maintain assets and report.
CWC and construction partners	Develop the proposal and tender delivery packages. Any construction margin is separate from shareholder equity IRR.
Equity investors	Fund $625.44m base contributions, bear equity risk and approve documented support commitments.
Senior lenders	Fund $800m subject to diligence; set covenants, account controls and completion tests.
Operations / terminal companies	Operate locks, water systems, terminals and maintenance under performance contracts.
Independent reviewers	Verify costs, hydrology, traffic, safety, model, legal powers and acceptance evidence.

The authority is a placeholder for a legally constituted public sponsor, not an assumed new federal agency. CWC is a concept brand, not a represented incorporated contractor or a licensed investment arranger.

Concession economics must state who bears freight demand risk, water-supply costs, extraordinary change in law, force majeure, unavailable navigation and major asset renewal. Do not place an uncontrollable risk in a private contract and presume that the public will never pay for it.

Published USDOT P3 guidance recognizes contractual allocation of delivery, financing, operating and sometimes traffic-revenue risks. A PPP is a delivery structure, not a source of free money. [3]

11 / FUNDING OPTIONS

Use federal programmes only where eligible

Instrument	Proposed use / limitation
Institutional equity	Long-duration infrastructure capital, with returns tested against contractual risk; no assumed investor appetite.
Commercial debt / private placement	Debt sized to CFADS, resilience and contract quality, not simply a target percentage of capital cost.
Public capital funding	The $2bn is a generic required contribution. No federal, state or local programme is named as an awarded source.
Availability payments	A budgeted operating obligation of the procuring authority, subject to enforceability and affordability review.
TIFIA	Potentially relevant to eligible surface-transportation / intermodal port elements. A standalone canal is not presumed eligible; up to 49% is not an entitlement. [4]
Marine Highway grants	Potentially relevant eligible components on designated routes; programme scale and awards cannot be assumed to fund this multibillion-dollar gap. [5]
RRIF / tax-exempt financing	RRIF relates to eligible railroad projects; no canal eligibility, tax-exempt bond qualification or tax benefit is assumed. [4]

Engage the relevant agencies early for a written eligibility determination. Do not double-count the same eligible cost under multiple programmes or assume that federal assistance eliminates matching, environmental, domestic-content, credit or other applicable requirements.

TIFIA material identifies creditworthiness and dedicated repayment requirements. The modeled private debt remains a conventional illustrative loan at 5.5%; no subsidized federal interest rate or guarantee is embedded. [4]

12 / DOWNSIDE AND INVESTOR RISK

Overruns reduce returns even if debt stays covered

Scenario	Equity IRR	Min DSCR
Base	12.00%	1.74x
Freight volume -25%	10.15%	1.55x
Opex +20%	11.25%	1.62x
Availability deduction -10%	10.97%	1.57x
Construction cost +20%	7.44%	1.74x
COD debt rate 7.5%	10.78%	1.42x
Opening delayed 1 year	10.66%	1.74x
Combined stress	3.96%	1.01x

Scenario definitions

Freight downside reduces both toll and handling volume by 25%, but leaves ancillary income unchanged. Opex stress raises the $35m base by 20%, leaving the $12m renewal provision unchanged. Availability stress cuts each annual payment by 10%.

Cost stress adds 20% of $3,231m non-financing construction cost ($646.2m) at the end of construction, funded entirely by equity. COD-rate stress raises only the operating debt rate to 7.5%, with a higher funded DSRA; construction pricing is unchanged. Delay adds one year of $44m debt interest and $15m holding cost, paid by equity, then shifts the full operating term one year.

Combined stress applies the freight, opex, availability, cost and COD-rate cases together; it excludes the delay. Positive residual cash is trapped when DSCR is below 1.30x. The 1.01x combined minimum falls below that threshold and may breach negotiated loan terms. Its reported IRR is a model output, not evidence that lenders would permit the case.

No probability weights are assigned, and no reserve sufficiency, default waiver or sponsor support is assured. These sensitivities are not a full stochastic forecast.

13 / RISK ALLOCATION AND MITIGATION

Make completion and water risks explicit

Risk	Commercial response to test
Water availability / drought	Secure lawful supply and operating plan; model restrictions and recurring pumping costs; do not finance a dry canal on assumed allocations.
Geology / land / utilities	Survey before bid; quantify exclusions and change mechanisms; maintain realistic contingency and public interfaces.
Construction completion	Independent engineer, cost-to-complete tests, contractor security and funded overrun support; avoid unconditional fixed-price claims.
Traffic and tariff acceptance	Independent demand analysis; shipper contracts; logistics-cost comparison; limited credit for speculative property income.
Availability / public credit	Confirm authority, budget treatment, payment security, deductions and dispute process; no automatic federal backing.
Inflation / interest rates	Match contract indexation and costs; test hedges and funding availability rather than assume perpetual cheap capital.
Operations and renewals	Condition surveys, validated renewal plan, adequate staffing, cyber safeguards and emergency response.
Handback / termination	Agree asset condition and compensation formulas; test termination exposures and lender rights without promising full repayment.

Water and environmental gates must precede financial close. A required payment cannot make an unsuitable mountain or desert alignment feasible. An investor return supported by payments also cannot substitute for legal authority, rights, ecological review or an acceptable public outcome.

14 / PUBLIC VALUE AND PROCUREMENT

Do not select a canal merely because IRR is positive

Commission a public-sector comparator and an independent economic appraisal covering the same freight movements and service quality. Compare canal improvement with rail capacity, road logistics, coastal services, terminal upgrades and the do-minimum case. Use the applicable public appraisal method and rates at the decision date. [2]

Test	Required evidence
Public affordability	Budget can support $2bn capital and the escalating payment without displacing higher-value obligations.
Economic case	Quantified freight savings, reliability, safety and environmental effects, less real resource costs and externalities.
No double counting	Tolls and public subsidies are transfers between parties, not additional social benefits simply because they appear as SPV revenue.
Value for money	Compare whole-life cost, financing premiums, risk transfer, flexibility, procurement and monitoring costs.
Competition	Tender comparable service requirements; disclose CWC related-party fees; test alternative designs and support packages.
Distributional impacts	Assess communities, landowners, Tribes, water users, public access and environmental burdens.

The 7% nominal discount rate used for financial and fiscal illustrations here is selected for sensitivity, not prescribed as an OMB economic-appraisal rate. Real and nominal rates must not be mixed. A public project can justify support through real external benefits, but those benefits must be demonstrated rather than inferred from this financing exercise.

If a lower-cost alternative meets the transport objective, do not raise availability payments merely to preserve a desired canal equity return.

15 / DEVELOPMENT PLAN

Release money only as evidence improves

Gate	Action / decision evidence
1. Sponsor and scope	Establish legal sponsor, corridor purpose, preliminary connection and existing asset interfaces.
2. Feasibility	Obtain route surveys, geotechnical evidence, lawful water-supply options, traffic and terminal studies.
3. Commercial structure	Test tariffs and shippers; procure candidate delivery options; define concession obligations and risk allocation.
4. Public business case	Independent appraisal, comparator, affordability, public and Tribal engagement, approvals strategy.
5. Bankability	Detailed model, tax advice, cost estimate, committed grants, contract terms, lender diligence and equity support.
6. Financial close	Executed funding, permits, land / water rights, contractor arrangements, insurances and reserves.
7. Delivery and opening	Certified progress, draw controls, cost-to-complete reports, acceptance tests and published navigation conditions.

No construction start or delivery date is promised. The five-year build period is a modeling assumption. Early development work should be budgeted and authorized separately, with a stop decision if evidence weakens.

The initial investment request

Invite qualified parties to assess a corridor development programme and fund a bounded feasibility stage on negotiated terms. Do not solicit subscriptions against the illustrative 12% return. The modeled $625.44m equity and $800m debt become a financing ask only after due diligence, negotiated public support and approvals.

16 / INDICATIVE INVESTMENT TERMS

A reviewable negotiation framework

Proposed item	Discussion position
Investment vehicle	Ring-fenced corridor concession SPV; jurisdiction, ownership and powers to be determined.
CWC position	Development / delivery sponsor; no exclusive contractor appointment presumed.
Equity amount	$625.44m base; additional cost and completion support to be negotiated.
Return objective	12% nominal pre-tax equity IRR; not a guaranteed or minimum distribution.
Public support	$2bn construction contribution plus solved opening payment of $96.16m, escalated 2%, subject to procurement and affordability.
Senior facility	$800m construction / operating financing; 5.5% assumed operating coupon; 30-year amortization.
Investor controls	Reserved matters, audited accounts, draw tests, related-party transaction approval and independent technical review.
Distribution policy	Debt and reserve tests first; base model uses a 1.30x DSCR distribution threshold.
Exit / refinancing	No resale proceeds, valuation uplift or refinancing dividend assumed. Any future transaction requires consent and model revision.
Handback	Zero sale / residual asset value to equity in the model; required condition and funding to be agreed.

This is a business concept, not an offering memorandum, securities prospectus, credit rating, underwriting commitment or regulatory approval. Any real capital raising requires the appropriate legal and regulated parties, verified disclosures and definitive agreements.

APPENDIX A / MODEL MECHANICS

Reproduce the return calculation

Dollar values in the model are US$m. The annual timeline begins at financial close, t = 0. All costs are nominal planning allowances; they are not a survey-based escalated estimate. The annual operating escalation factor is 1.02 raised to (operating year minus 1).

Calculation	Definition / result
Freight volume	12 x ramp; ramp = 0.60, 0.80, 0.90, then 1.00
Commercial income	(6 x freight volume + 6 x ramp) x escalation factor
Public income	96.1590401265 x escalation factor; no deductions in base
CFADS	Commercial + public income - 35 x factor - 12 x factor
Debt payment	800 x 0.055 / [1 - (1.055)^(-30)] = 55.0443117435
Debt interest	Opening principal x 5.5%; principal repaid = payment - interest
Equity cash flow	CFADS - debt service + reserve / working-capital releases, subject to modeled cash trap
Construction interest	Sum of (opening debt + half annual draw) x 5.5% = 103.4
IRR condition	Sum of annual cash flow / (1 + r)^year = 0
Availability solution	Vary opening payment until equity NPV at 12% equals zero

The DSRA is funded in build year 5 and released after scheduled debt repayment in operating year 30. Working liquidity of $20m is returned in year 40. Annual renewal contributions are assumed consumed / committed; no accumulated renewal balance is released. There is no terminal asset valuation.

Independent model checks confirm that construction sources equal uses, scheduled principal repayments total $800m and the base equity NPV residual at 12% is effectively zero. This is internal arithmetic validation, not an independent audit or validation of commercial assumptions.

APPENDIX B / OPENING CASH-FLOW BRIDGE

From revenue to distributions

US$m unless stated	Op year 1	Op year 4	Op year 30	Op year 40
Commercial receipts	46.80	82.77	138.52	168.85
Availability payment	96.16	102.04	170.76	208.16
Operating cost	35.00	37.14	62.15	75.77
Renewal provision	12.00	12.73	21.31	25.98
CFADS	95.96	134.94	225.81	275.27
Debt interest	44.00	42.08	2.87	0.00
Debt principal	11.04	12.97	52.17	0.00
Total debt service	55.04	55.04	55.04	0.00
DSRA release	0.00	0.00	55.04	0.00
Working-capital release	0.00	0.00	0.00	20.00
Equity distribution	40.91	79.90	225.81	295.27

Year 30 distribution includes the $55.04m DSRA release after debt repayment. Year 40 includes the $20m working-capital release. Commercial receipts and public payments shown are nominal amounts in each year, not opening-price figures.

Construction funding is detailed in Section 04. For gross asset IRR, use $3,231m non-financing construction expenditure over the same five-year profile plus $20m working capital in year 5. Exclude financing fees, construction interest, DSRA, grants and debt flows; include availability only in the supported asset case.

State / local and federal taxes, depreciation, tax credits, investor fees, contractor default, changes in working capital, actual inflation paths, grant clawbacks, unexpected asset renewals and litigation are not modeled. Incorporating these may materially reduce returns or increase support needs.

APPENDIX C / ANNUAL OPERATING CASH FLOW 1-20

Operating years 1-20

US$m, nominal and pre-tax. Total project year = five construction years + operating year. Rounded display; calculations use unrounded values.

Op yr	Project yr	Commercial	Public pay	CFADS	Debt pay	Equity cash	Debt close
1	6	46.80	96.16	95.96	55.04	40.91	788.96
2	7	63.65	98.08	113.79	55.04	58.75	777.30
3	8	73.04	100.04	124.18	55.04	69.14	765.01
4	9	82.77	102.04	134.94	55.04	79.90	752.04
5	10	84.43	104.09	137.64	55.04	82.60	738.36
6	11	86.12	106.17	140.39	55.04	85.35	723.93
7	12	87.84	108.29	143.20	55.04	88.16	708.70
8	13	89.60	110.46	146.07	55.04	91.02	692.63
9	14	91.39	112.67	148.99	55.04	93.94	675.68
10	15	93.22	114.92	151.97	55.04	96.92	657.80
11	16	95.08	117.22	155.01	55.04	99.96	638.94
12	17	96.98	119.56	158.11	55.04	103.06	619.03
13	18	98.92	121.95	161.27	55.04	106.22	598.03
14	19	100.90	124.39	164.49	55.04	109.45	575.88
15	20	102.92	126.88	167.78	55.04	112.74	552.51
16	21	104.98	129.42	171.14	55.04	116.10	527.86
17	22	107.08	132.01	174.56	55.04	119.52	501.84
18	23	109.22	134.65	178.05	55.04	123.01	474.40
19	24	111.40	137.34	181.61	55.04	126.57	445.45
20	25	113.63	140.09	185.25	55.04	130.20	414.90

Commercial = tolls + terminal handling + ancillary income. CFADS deducts operating and renewal costs. Equity cash includes the DSRA release in operating year 30 and working liquidity in year 40. There is no distribution cash trap in the base case because DSCR remains above 1.30x.

APPENDIX C / ANNUAL OPERATING CASH FLOW 21-40

Operating years 21-40

US$m, nominal and pre-tax. Total project year = five construction years + operating year. Rounded display; calculations use unrounded values.

Op yr	Project yr	Commercial	Public pay	CFADS	Debt pay	Equity cash	Debt close
21	26	115.90	142.89	188.95	55.04	133.91	382.68
22	27	118.22	145.75	192.73	55.04	137.69	348.68
23	28	120.59	148.66	196.59	55.04	141.54	312.82
24	29	123.00	151.63	200.52	55.04	145.47	274.98
25	30	125.46	154.67	204.53	55.04	149.48	235.05
26	31	127.97	157.76	208.62	55.04	153.57	192.94
27	32	130.53	160.91	212.79	55.04	157.75	148.51
28	33	133.14	164.13	217.05	55.04	162.00	101.63
29	34	135.80	167.42	221.39	55.04	166.34	52.17
30	35	138.52	170.76	225.81	55.04	225.81	0.00
31	36	141.29	174.18	230.33	0.00	230.33	0.00
32	37	144.11	177.66	234.94	0.00	234.94	0.00
33	38	146.99	181.22	239.64	0.00	239.64	0.00
34	39	149.93	184.84	244.43	0.00	244.43	0.00
35	40	152.93	188.54	249.32	0.00	249.32	0.00
36	41	155.99	192.31	254.30	0.00	254.30	0.00
37	42	159.11	196.15	259.39	0.00	259.39	0.00
38	43	162.29	200.08	264.58	0.00	264.58	0.00
39	44	165.54	204.08	269.87	0.00	269.87	0.00
40	45	168.85	208.16	275.27	0.00	295.27	0.00

Commercial = tolls + terminal handling + ancillary income. CFADS deducts operating and renewal costs. Equity cash includes the DSRA release in operating year 30 and working liquidity in year 40. There is no distribution cash trap in the base case because DSCR remains above 1.30x.

APPENDIX D / SOURCES AND PROVENANCE

Separate evidence from model assumptions

[1] American Canal Network: Proposed National Standard, independent concept report, 8 November 2026
Open source - Source of ACN-100 concept dimensions; not a government-issued standard.

[2] OMB Circular A-94: Guidelines and Discount Rates for Benefit-Cost Analysis of Federal Programs
Open source - IRR definition, role of NPV and limitations of IRR as a public decision criterion.

[3] USDOT Build America Bureau: Public-Private Partnerships
Open source - Published P3 delivery and risk-allocation context.

[4] USDOT Build America Bureau: TIFIA Credit Program Overview and Ports
Open source - Eligibility, dedicated repayment and credit assistance limits; no eligibility or award assumed.

[4a] USDOT Build America Bureau: Ports
Open source - Port-related TIFIA and railroad-related RRIF context.

[5] USDOT: United States Marine Highway Program, FY2026 overview
Open source - Competitive programme and eligible-component context; not a source of an awarded $2bn grant.

Assumption provenance

Prior CWC concept: $150m/open mile, $250m/lock, $60m/bridge and 35% corridor allowance. All additional costs, $2bn public contribution, 5.5% debt rate, 12% hurdle, 2% escalation, revenue and cost assumptions, 7% financial discount sensitivity, 40-year operations and all stress cases are analyst inputs. They are not market quotes, forecasts, lender terms, awarded grants or investor commitments.

Sources checked 8 November 2026. No present-day programme availability is promised. Cover image: AI-generated American canal design concept from this conversation. Report and financial calculations prepared as a standalone business proposal for discussion.